

**UCSC****University of Colombo, Sri Lanka***University of Colombo School of Computing***DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**Academic Year 2023 — 3rd Year Examination — Semester 6**IT6306 — Mobile Application Development***Structured Question Paper*
(2 Hours)**To be completed by the candidate****Index Number**

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Important Instructions

- The duration of the paper is **2 hours**.
- The medium of instructions and questions is English. Students should answer in the medium of English language only.
- This paper has **4 questions** on **10 pages**. Answer **all** questions.
- All questions carry **equal** marks.
- Write your answers **only on the space provided** on this question paper.
- Do not tear off any part of this question paper. Under no circumstances may this paper (or any part of this paper), used or unused, be removed from the Examination Hall by a candidate.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigilator immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Non-programmable calculators are **not allowed**.
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**To be completed by
the examiners**

1	
2	
3	
4	
Total	

1. (25 marks)

- (a) Name two (02) mobile Operating Systems.

(2 marks)

Android, iOS, Symbian, etc.

- (b) What is the use of the Operators layer in the mobile ecosystem?

(3 marks)

The Operators layer in the mobile ecosystem manages the network infrastructure, providing mobile connectivity for calls, texts, and internet access. They issue SIM cards, handle customer accounts, and offer additional services like international roaming and emergency call support. Operators also ensure network security, compliance with regulations, and continuous technological improvements.

- (c) What is the difference between the "Application Frameworks" and the "Applications" layers of the Mobile Ecosystem? Bring three (03) points to gain full marks.

(6 marks)

The Application Frameworks layer provides the tools and libraries developers use to build mobile applications, offering essential functions like user interface components, data management, and hardware access. The Applications layer consists of the apps that users interact with, such as social media, games, and productivity tools. These are built using the frameworks provided by the Application Frameworks layer.

- (d) What is the difference between the Second Generation (2G) and Third Generation (3G) mobile wireless networks? Bring three (03) points to gain full marks.

(6 marks)

The technical difference between Second Generation (2G) and Third Generation (3G) networks lie in their underlying technologies and data capabilities. 2G networks, based on GSM or CDMA, provide basic digital voice and limited data services at speeds up to 64 Kbps. In contrast, 3G networks, utilizing technologies like UMTS or CDMA2000, offer significantly faster data speeds, up to 2 Mbps or more, enabling advanced services such as mobile internet, video calls, and multimedia streaming.

- (e) Elaborate and give two (02) real-world examples for the Mobile Strategy rule: "You cannot support everything."

(8 marks)

The Mobile Strategy principle "You cannot support everything" underscores the importance of prioritization and focus in mobile development. It acknowledges that attempting to cater to every platform, feature, or user preference can dilute resources and compromise the quality of the product. For example, Snapchat initially focused on iOS to perfect its user experience before expanding to Android, ensuring a polished product launch and effective user engagement. Another instance is Slack, which concentrated on desktop platforms before adapting its messaging app for mobile, aligning its strategy with user needs and technological capabilities. By strategically choosing where and how to invest resources, companies can optimize development efforts and deliver superior mobile experiences that resonate with their target audience.

02.

(25 marks)

(a) Name two (02) mobile application monetization strategies.

(2 marks)

Free and Paid App Versions Model

Free App with In-app Purchases Model

Free App with Subscription Model

Paid App Model

Partnership Model (any 2)

(b) Explain the use of the eCPM (Effective Cost Per Mille) metric in Google AdMob Metrics.

(3 marks)

The eCPM (Effective Cost Per Mille) metric in Google AdMob measures the estimated earnings generated for every 1000 ad impressions an app serves. It calculates the average revenue earned per thousand impressions, regardless of the ad format (e.g., banner, interstitial, rewarded video). This metric helps app developers and publishers understand the overall revenue potential of their ad inventory and compare the performance of different ad units or ad networks.

(c) Why do we need to sign a mobile application when we publish on app stores?

(4 marks)

Signing a mobile application before publishing it on app stores is crucial for ensuring its Authenticity, integrity, and security. By digitally signing the app, developers provide a verified source, protecting users from potential tampering or malware. Additionally, app store compliance mandates signing applications, streamlining the review process, and ensuring adherence to store guidelines.

- (d) If you want to ensure that your mobile application is accessible to most people worldwide, which platform would you choose? Justify your answer.

(4 marks)

Choosing the Android platform ensures maximum worldwide accessibility due to its dominant market share, especially in emerging markets, diverse device availability, extensive language support, and broader reach through the Google Play Store. Its open-source nature and affordability make it accessible to users with varying economic backgrounds, allowing developers to reach a wider audience globally.

- (e) What is the difference between "Cocoa Touch" and "Media" Layers in the iOS Layered Architecture? Bring three (03) points to gain full marks.

(6 marks)

In the iOS Layered Architecture, the "Cocoa Touch" layer is the foundational framework for building iOS applications, providing essential components and functionalities for user interface design, event handling, and device hardware access. In contrast, the "Media" layer manages multimedia content within applications, offering audio and video playback, graphics rendering, animation, and multimedia processing frameworks. While Cocoa Touch focuses on general application development and user interaction, the Media layer handles multimedia assets, enabling developers to seamlessly integrate audio, video, and graphics into their iOS applications, enhancing the overall user experience.

- (f) How does the user's physical environment impact their overall experience with a mobile application? Explain using two (02) real-world scenarios.

(6 marks)

The user's physical environment significantly impacts their experience with a mobile application, influencing usability, performance, and overall satisfaction. For instance, using a navigation app in a crowded urban area with tall buildings can result in GPS signal interference and a cluttered interface, hindering navigation efficiency and user satisfaction. Similarly, browsing a shopping app in a large department store with inconsistent Wi-Fi connectivity may lead to frustration and difficulty finding desired products due to inaccurate location tracking and search functionality.

03. (25 marks)

- (a) Briefly explain the two ways to register a broadcast receiver in an Android application. Also mention any alternative names used to denote these two types.

(6 marks)

1. Static Receiver (3 marks)

A static receiver is a broadcast receiver that is registered in the AndroidManifest.xml file.

This type of receiver is also known as a Manifest-declared receiver.

2. Dynamic Receiver (3 marks)

A dynamic receiver is a broadcast receiver that is registered at runtime using the application's or an activity's context in the Java files. This type of receiver is also known as a Context-registered receiver.

L4 – Slide 79

- (b) Describe the similarity and differences of the steps of initiating and ending a service's life-cycle in Android environment, specifically the service types: started service, bound service and intent service.

(6 marks)

Started Service Lifecycle Management: A started service must manage its own lifecycle. If it is

not stopped, it will keep running and consuming resources. The service can stop itself by calling the `stopSelf()` method. This method ensures that the service ceases execution and releases any resources it was using.

An activity or another service, can stop a started service by calling the `stopService()` method.

This stops the service if it is currently running. A bound service, which is bound to one or more clients, is destroyed when all clients unbind from it. Once all clients have disconnected, the service is no longer needed and is destroyed. An `IntentService` is automatically destroyed after its `onHandleIntent()` method returns. This ensures that the service completes its task and then shuts down, releasing its resources.

L4 - Slide 91

(c) There are three categories that Android sensors are grouped into.

i) Describe each of them briefly with one (01) example for each category.

(6 marks)

ii) State two (02) features of the SensorManager component in Android sensor framework.

(2 marks)

i) Motion Sensors: Motion sensors measure acceleration forces and rotational forces along three axes. Ex: An accelerometer/Gravity sensor/Gyroscope/Rotational vector sensor

Environmental Sensors: Environmental sensors measure various environmental parameters, such as ambient air temperature, pressure, illumination, and humidity.

Ex: A barometer/Photometers (light sensors)/Thermometer

Position Sensors: Position sensors measure the physical position of a device.

Example: A magnetometer/Proximity sensors

L4 – Slide 105-106

ii) Access and listen to sensors, Register and unregister sensor event listeners

Acquire orientation information, Provides constants for accuracy, data acquisition rates, and calibration

L4 – Slide 108

- (d) Write a POJO class named BITStudent to represent the following properties in Java: an int type property called sid, a String type property called fullName, a boolean type property called isOnline. The class should be 'public' and all 3 properties should be 'private'. Using the appropriate syntax, annotate the above POJO class such that its instances can be persisted via the Room persistence framework.

Note: sid will be a primary id and it should be assigned automatically. The sid and fullName will be represented in columns named sid and fname, respectively. The online state is a runtime-only property and must never be persisted. Additional methods or constructors are not required.

Ignore the necessity of the respective import statements as well.

(10 ÷ 2 = 5 marks)

@Entity(tableName = "bit_students") //1 mark for @Entity (tableName ... not given in the note)

public class BITStudent {

@PrimaryKey(autoGenerate = true) // 3 marks

private int sid; //1 marks

@ColumnInfo(name = "fname") // 3 marks

private String fullName; // 1 mark

private boolean isOnline; // 1 mark

}

L4 – Slide 108

04.

(25 marks)

- (a) Describe the importance of using design tools for mobile application development. Elaborate on four (04) distinct aspects where these tools play a critical role, contributing to a positive flow of the process. Each point should be clearly numbered in your response.

(2 x 4 = 8 marks)

Speed up the design process (Quickly create and modify prototypes)

Improve collaboration (Real-time sharing, commenting, and version control)

Enhance user experience (Refine designs based on user feedback)

Cost-effective (Save time and resources by iterating before development)

Describing each of the above will get the full marks.

Other valid points with description is accept

Lecture 3 Slide 70

- (b) Explain the following mobile application design concepts briefly and give a concrete example for each of them: Context, Typography, Look and Feel, and Message. **(2 x 4 = 8 marks)**

Context : The environment, circumstances, and conditions surrounding users when they interact with a mobile app or website. The designer may become aware of the level of light in a given outside condition and increase the screen brightness to make it visible to the user. Slide 12-to-15

Typography : Typography is the technique of arranging type to make written language legible, readable and appealing when displayed. One example of applying different type faces, eg: Sans/Sans serif/Comic fonts according to the need. Slide 46,53

Look and feel refers to the appearance and style of a design. Look and feel of an Android app may be different from the iOS app as they inherit them from the OS. Mixing the two would be unsuitable.

Message : Message is the overall impression you create through (visual) design. A children's movie application may use lot of eye catching visual information and action character to give the appropriate message to the intended audience.

Lecture3 - Slide 17, slide 24

- (c) i) Briefly explain/define the concept of 'color depth' with respect to the mobile application design. **(2 marks)**
- ii) Calculate and find the number of bits required for 1024 shades and 4096 shades in a device. **(3 marks)**

i) Color depth refers to the number of bits representing the color of a single pixel in a digital

Image //2 marks

ii) 1024 shades = $2^{10} = 10$ bits

4096 shades = $2^{12} = 12$ bits

2 marks for correct answers 1 mark for calculation

Lecture4 Slide 35

- (d) In the layered architecture of Android platform, the 'user apps and system apps' are in the top layer.

Name four other major layer components in the Android platform architecture.

(1 x 4 = 4 marks)

HAL /Hardware abstraction layer (ANY four)

Java API Framework,Native C/C++ libraries

(Linux) Kernel, ART/ Android Runtime

Lecture4 Slide 8
